Preference-Aware Coefficient Correction for Rewarded-Soups-Style Model Merging

Motivation and Goal

$$\theta_i \rightarrow \delta_i = \theta_i - \theta_0 \rightarrow R \rightarrow \lambda^* = f(p, R)$$

Motivation:

User preferences p are defined over objectives, but model merging happens in task-vector space.

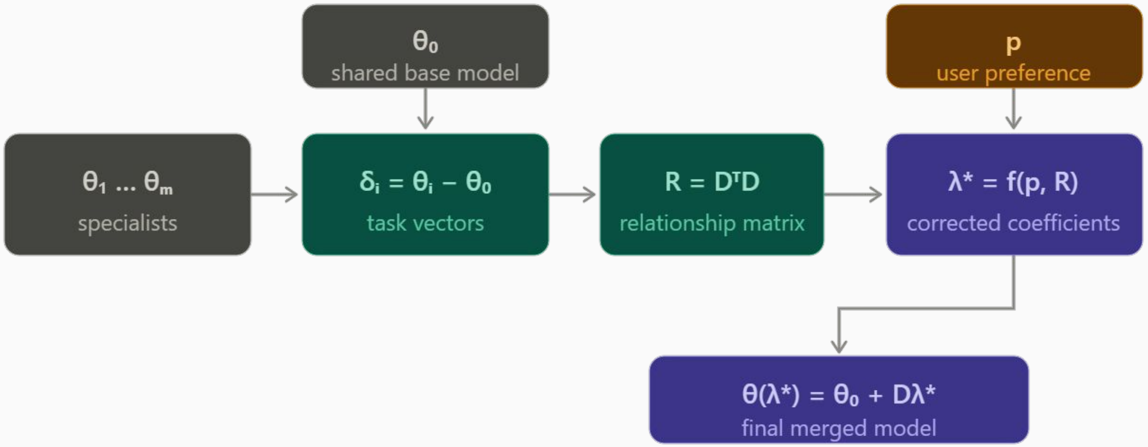
Goal:

Use task-vector geometry R to compute preference-aware merge coefficients λ^*.

Architecture Overview + Method

$$D = [\delta_1, \ldots, \delta_m]$$

- Base model: **TinyLlama-1.1B**
- Five HelpSteer2 LoRA specialists
- Compute relationship matrix: $R = D^\top D$
- Compute corrected coefficients: $\lambda^* = f(p, R)$
- Merge adapters and evaluate with **ArmoRM**


θ0
shared base model
p
user preference
θ1 … θm
specialists
δi = θi − θ0
task vectors
R = DᵀD
relationship matrix
λ* = f(p, R)
corrected coefficients
θ(λ*) = θ0 + Dλ*
final merged model

Adapter training setup

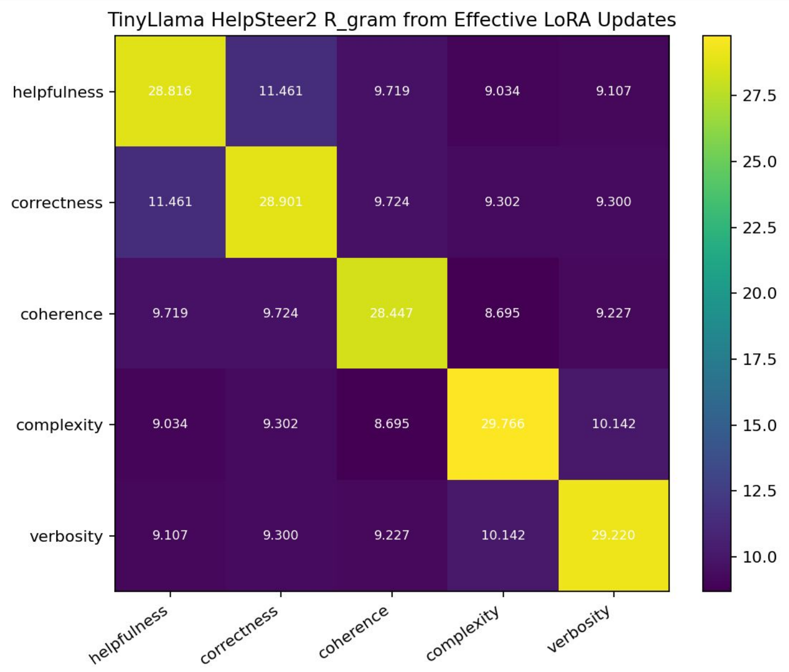
- Training examples are filtered from HelpSteer2 based on high attribute scores
- Response-only supervised fine-tuning
- Training is run adapter by adapter
- Best checkpoints are selected using evaluation loss

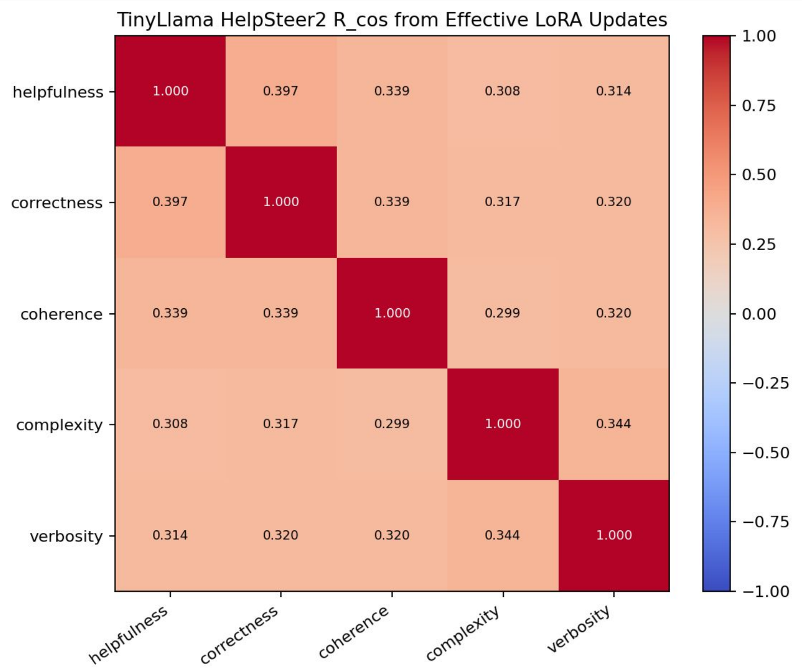
Training workflow improvements

- Switched to a cosine_decay learning-rate schedule
- Added token-length checks and increased max_length to 2048
- Added grad_norm tracking
- Added SAVE_BEST_BY_EVAL_LOSS

(Show graphs)

Relationship matrix R





Method Difficulties

- Some Methods used C(λ)

$$R^{-}_{ij} = \max(0, -R_{ij}) = 0$$

$$C(\lambda) = \lambda^{\top} R^{-} \lambda \equiv 0$$

λ*=p

- Consider further papers: FairGrad

Next Steps

- Run proxy validation to check assumption
  - The goal is to test whether geometric relationships between adapters are related to reward behavior
- Rework methods
- Update thesis text